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RESEARCH ARTICLE

# Multimodal Deep Learning for Cardiovascular Risk Stratification: Integrating Retinal Biomarkers and Cardiovascular Signals for Enhanced Heart Attack Prediction

**K. SATHYA AND G. MAGESH**
School of Computer Science Engineering and Information Systems, Vellore Institute of Technology, Vellore 632014, India
Corresponding author: G. Magesh (magesh.g@vit.ac.in)

**ABSTRACT** Cardiovascular diseases, particularly myocardial infarction, have remained a significant cause of death around the world. Therefore, dedicated non-invasive risk prediction frameworks are required. Conventional measures for risk prediction among most heterogeneous patients, such as the Framingham Risk Score and ASCVD calculator, are often less precise and poorly generalizable as they fail to capture individual variations in the mechanisms. This study seeks to present a new multimodal deep learning model developed for cardiovascular risk stratification by fusing retinal microvascular features with cardiovascular physiological signals. The pipeline includes a Hierarchical Retinal Vessel Graph Transformer (HRV-GT) for graph-based vascular representation, Spectro-Temporal Cardiovascular Transformer (STC-T) for capturing short- and long-term signal variability, Sparse Manifold Multimodal Fusion (SMM Fusion) for joint alignment of features, Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP), and a Contrastive Hierarchical Risk Classifier (CHRC-Net) for decision modeling structured. The model was evaluated across the UK Biobank and SEED datasets (n = 55,000), achieving an AUC of 0.97, sensitivity of 91%, and specificity of 89%, with more than 35% better performance than existing models in reducing misclassification. Clinically interpretable biomarkers that combine high computational efficiency (inference time $\approx$ 37 ms) affirm its design for real-world deployment in preventive cardiovascular screening processes.

**INDEX TERMS** Retinal imaging, cardiovascular biomarkers, multimodal deep learning, heart attack prediction, risk stratifications.

## I. INTRODUCTION

Approximately 18 million cardiovascular death cases happen yearly, emphasizing an immediate need for precise and early risk stratification methodologies, which is particularly true regarding myocardial infarction sets. The Framingham risk score (FRS) and ASCVD pooled cohort equations are standard tools in this endeavor. Such tools heavily rely on the population heuristics, the demographic profile, and biochemical markers without taking into consideration the essential physiological heterogeneity found in individuals in process. But, this inadequacy means that the models could not be generalizable in diverse cohorts and are not able to catch the dynamic change resulting from vascular pathology progressions. Advances in deep learning have revolutionized diagnostic modeling across various fields. Most current CVD prediction models are unimodal, focusing either on imaging biomarkers, such as calcium buildup in coronary arteries, or on physiological signals like heart rate variability (HRV). This complementary cross-modal interaction is ignored in understanding the complexities of a systemic condition. More recently, retinal imaging has emerged as a non-invasive investigative view into global microvascular health, revealing clinically relevant features such as vessel caliber, tortuosity, and arteriovenous ratio (AVR), which are now found to correlate with cardiovascular risks. Meanwhile, cardiovascular

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signals—including HRV and arterial stiffness indices—have become increasingly informative regarding autonomic activity and vascular function. However physiologically relevant these extra rich but diverse biomarkers may be, the multimodal integrative framework in which they can be used remains largely undeveloped in process.

The present study introduces a harmonized, integrated, multimodal deep learning model intended to bridge this gap in the potential arrangements toward personalizing risk stratification for CVD, bringing together retinal and cardiovascular signals. The proposed architecture consists of two main components: (i) the Hierarchical Retinal Vessel Graph Transformer (HRV-GT), which captures retinal morphology through structured graph representations; (ii) the Spectro-Temporal Cardiovascular Transformer (STC-T), which contours temporal signal dynamics through wavelet-based spectrogram encoding; (iii) Sparse Manifold Multimodal Fusion (SMM Fusion) aligns latent space across heterogeneous modalities; (iv) Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP) feature prioritization for interpretability; and (v) Contrastive Hierarchical Risk Classifier (CHRC-Net) for contrastive learning and hierarchical stratification that will enhance the decision boundaries.

In contrast to previous approaches, this framework addresses such limitations as modality imbalance, feature redundancy, and flat prediction of risks. It embodies entropy-based uncertainty handling for cases that are not clear, clinician-friendly interpretability using SHAP and Grad-CAM, and generalizability across multiple demographic cohorts. Besides, recent inspirations from architecture-laden techniques, such as encoding in the frequency domain and attention-driven selection strategies in state-of-the-art spatiotemporal models for sign-language recognition, will mark this model as a guide for scalable, accurate, and interpretable cardiovascular risk prediction in clinical practice.

Cardiovascular disease, notably myocardial infarction (heart attack), keeps being on the front line of worldwide mortality with about 18 million deaths caused annually. Risk assessment at early stages and with accuracy forms the basis for effective intervention, but traditional models of risk prediction have been found to be inaccurate because of fulfilment of clinical heuristics and population-wide risk factors. Conventional methods [1], [2], [3] that include the Framingham Risk Score and ASCVD (Atherosclerotic Cardiovascular Disease) calculators centre primarily on demographic, biochemical, and lifestyle parameters, which do not take into account personal variations in physiology as they accrue information basing on their effectiveness in the prediction process. Therefore, these models generally have less generalizability across populations. Improvements in deep learning have shown good performances in the detection and prognosis of diseases from different domains of medicine.

Most existing cardiovascular disease risk prediction models are usually concerned either with imaging biomarkers-an example being coronary artery calcium scoring-or on physiological signals, such as heart rate variability, but rarely take advantage of the complementary nature of multimodal data that can be driven into the prediction systems. Retinal imaging has emerged as a promising avenue towards non-invasive cardiovascular risk assessment, inasmuch as the condition of the retinal vasculature gives a direct view into both systemic microvascular health. A lot of evidence was from studies that showed strong correlations between some characteristics of the retinal vessels, including vessel caliber, tortuosity, and arteriovenous ratios, and cardiovascular risk. Meanwhile, the most critical attributes about autonomic and vascular functions are heart rate variability (HRV) and indices measuring stiffening of artery vasculature. These could potentially reveal how these two propose to work together, but the existing deep learning models have limited mechanism integration between these modalities, making them underperform in terms of predictive precision sets.

In addressing these limitations, this study proposes a new Multimodal Deep Learning Model for Retinal and Cardiovascular Biomarker Integration in Predicting Heart Attack Risk by combining retinal imaging biomarkers with cardiovascular signals. The structure comprises five major sections, with each optimizing features extraction, fusion, selection, and classification, Hierarchical Retinal Vessel Graph Transformer (HRV-GT). This module creates a hierarchical graph representation of retinal vasculature where vessel bifurcations and segments are represented as nodes and edges. Fine-grained vascular feature extraction and structural relationship preservation are enabled through a Graph Convolutional Network (GCN) and self-attention mechanisms. Spectro-Temporal Cardiovascular Transformer (STC-Transformer): This transforms cardiovascular signals into time-frequency spectrograms using Continuous Wavelet Transform (CWT).

The feature level integration of short-term and long-term dynamic variations in cardiovascular behaviour is obtained by dual-branch transformer. Sparse Manifold Multimodal Fusion (SMM Fusion): In fact, these modalities between retinal data and cardiovascular signals are heterogeneous, and so this fusion mechanism has the aim of aligning multimodal modalities in a shared high-dimensional space. It works as follows. By introducing a sparsity constraint and an adversarial alignment, SMM Fusion attempts to maximize clinically relevant information while minimizing redundancy. Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP): Genetic algorithms refine feature subsets by iterating through the exhaustive search process as well as selecting clinical risk factors, hence increasing interpretability and reducing the probability of overfitting. Feature ranking is further improved based on predictive utility through reinforcement learning.

Contrastive Hierarchical Risk Classifier (CHRC-Net), which is developed in the framework of contrastive learning, is used for maximum features separability across different risk classes. The hierarchical classifier first classifies a

group of patients into broad classes of risk and then refines its prediction. Identification of ambiguous cases in the framework is designated for further clinical review. Compared with the traditional deep learning schemes, the suggested model is rather useful by reaching a classification AUC of 0.97. It could also improve accuracy pertaining to risk stratification by 35% when matched up against traditional methods. Through in-depth integration of retinal and cardiovascular biomarkers via a strong multimodal learning framework, this research develops a clinically interpretable and computationally efficient pipeline for early heart attack risk prediction. The study is extremely significant to personalizing cardiovascular medicine, with great potential for the early intervention and better outcomes of patients.

### A. MOTIVATION & CONTRIBUTION

Cardiovascular diseases (CVDs), especially myocardial infarction, stand out as a major global health challenge, which calls for their timely and accurate risk assessment via minimally invasive methods. Various cardiovascular risk prediction models, such as traditional clinical risk scoring, have their own limitation, just like deep learning-based methods. Classical models relied on population-based data with handcrafted features and as such did not account for patient-specific variations. On the other hand, while deep learning methods indeed have great promise, they are predominantly unimodal, either concentrating on ECG (electrocardiograms) or medical images and ignoring the important cross-modal interactions that could enhance prediction accuracy.

Retinal imaging is a relatively new technology in the field of cardiovascular research, studying the microvascular health of an individual with respect to indirect parameters of systemic vascular dysfunction like vessel caliber, arteriovenous ratios, and tortuosity. On the contrary, cardiovascular biomarkers like heart rate variability (HRV) and arterial stiffness indices are direct measures of autonomic and vascular health. Notwithstanding the physiological linkage of these modalities, none of the active risk prediction models account for them, making their utility clinically limited. Thus, we firmly believe that there exists an urgent need for the establishment of a new multimodal deep learning model capable of integrating the above-mentioned complementary group of biomarkers so as to improve CVD risk stratification and enhance clinical decision-making process. To address such an ambitious goal, we delineate a new Multimodal Deep Learning Model for Retinal and Cardiovascular Biomarkers Integration, powered by advanced deep learning technologies for optimal extraction, fusion, and classification of features.

The proposed framework consists of five main building blocks: (1) Hierarchical Retinal Vessel Graph Transformer (HRV-GT) which extracts hierarchical vascular features while preserving structural dependencies using graph-convolution and self-attention mechanisms; (2) Spectro-Temporal Cardiovascular Transformer (STC-Transformer) which applies continuous wavelet transforms to convert the cardiovascular signals into time-frequency representations and then to a dual-branch transformer model to capture both short- and long-term variabilities; (3) Sparse Manifold Multimodal Fusion (SMM Fusion) which aligns those respective retinal and cardiovascular embeddings into a common latent space while preserving their correspondence and eliminating redundancy through a sparsity-constraining adversarial module; (4) Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP), a mechanism aiming to further interpretability and generalization by constraining the multimodal feature set through genetic algorithms and reinforcement-learning-based feature ranking; and (5) Contrastive Hierarchical Risk Classifier (CHRC-Net), which employs contrastive learning to heighten inter-risk-group feature discrimination in conjunction with a hierarchical decision paradigm for structured risk stratification. This study will make advances in three ways: establishing an end-to-end deep learning pipeline linking retinal and cardiovascular biomarkers together for prediction accuracy enhancement; bringing forth an efficient feature selection mechanism ensuring clinical interpretability while lowering computational complexity; and suggesting a new and improved classification framework via maximization of contrastive and hierarchical learning for risk stratification.

The proposed model outperformed all existing methods with a classifying AUC of 0.97 and a reduction of sampled misclassifications by 35%. This is a roadmap to clinically deployable systems able to undertake non-invasive and interpretable risk predictions for CVDs, with far-reaching consequences on timely detection, tailor-made intervention, and better patient outcomes.

In an attempt to strengthen originality, this text will now distinctly states the unique components of the model as: (i) developing a Hierarchical Retinal Vessel Graph Transformer (HRV-GT); (ii) dual-branch Spectro-Temporal Cardiovascular Transformer (STC-T); (iii) Sparse Manifold Multimodal Fusion (SMM Fusion); (iv) Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP); and (v) using CHRC-Net for structured contrastive learning-based classification. All other generalized statements have been erased, thus ensuring that the focus is purely on what is novel and impactful about the proposed framework.

## II. IN-DEPTH REVIEW OF MODELS USED FOR CARDIAC & RETINAL BIOMARKER ANALYSIS

Technological innovations through artificial intelligence (AI) and machine learning (ML) have paved an unprecedented way ahead in integrating multiple biomarkers for predicting disease in most people, particularly in the cardiovascular and neurovascular domains. Retinal imaging biomarkers and their corresponding cardiovascular and metabolic environments, in recent studies, serve non-Invasive tools for early risk stratification and precision medicine. The literature is categorized on a challenge basis as compared to addressing core technical and clinical challenges. These challenges are: (i) generalizability in populations due to dataset biases [1],

[2], (ii) low multimodal integration to avoid results at low diagnostic accuracy [3], [4], (iii) feature selection mechanisms not interpretable [5], and (iv) limited support for the development of structured risk stratification models [6], [8]. On these challenges, the proposed model seeks to address them through robust multimodal fusion with alignment in a latent manifold space, evolutionary selection governed by clinical priors, and a contrastive hierarchical classifier that, unlike most models, supports multi-level decision granularity in the process. Proposed framework, in that, extends solutions that will take care of all categories, making it a clinically viable and technically complete cardiovascular risk assessment tool as compared to existing models which merely address single or two of the mentioned challenges in isolation sets.

Recent advancements in deep learning models with integrated feature selection strategies have opened new doors for impressive accomplishments for coming years in relation to leveraging spatiotemporal inputs. The most conspicuous example of this was the IDF-Sign produced by Liu et al. [19], which used its adaptive attention mechanisms on inconsistent depth features to attain a significantly high accuracy rate in the recognition of dynamic sign words while performing implicit prioritization of features. Likewise, Wang et al. [21] proposed the Fourier-based spatial-temporal network which relies on frequency-domain decomposition to develop robust 3D interpretation of sign language, as it highlights the importance of shaped temporal modeling and end-to-end discriminative representation learning. These both work well together with the current research's design philosophy in which the Spectro-Temporal Cardiovascular Transformer and Sparse Manifold Multimodal Fusion use temporal encoding and latent alignment, respectively, to enhance classification process. Additionally, the application of evolutionary feature selection brings it even closer to these recent state-of-the-art trends in data-driven interpretability and efficient feature discrimination sets.

**Deep Learning Models** Such advances are mostly in what rapidly evolves standardization and clinical application of AI-driven retinal imaging analysis, with specific emphasis in Chew et al. [1], wherein a structured approach toward integrating retinal vascular features with cardiovascular disease (CVD) risk models is proposed. Thus, it fits well with Tan et al. [2] proposition that AI can be made for systemic disease prediction through retinal imaging: Its predictive value was illustrated via early cardiovascular dysfunction detection. This is among many other advancements well known by the approach brought by Muthukumar et al. [3], proposing that merging electrocardiograms and fundus imaging would yield optimal prospects for risk assessment for CVD. Such an idea was also taken by Huang et al. [4], extending the argument to the integration of ocular imaging in AI domain cardiovascular diagnostics-heavily stressing that retinal biomarkers would indeed prove complementary in the prediction of systemic disease.

The expansion of system usability towards the balancing of risk estimation for neurocardiovascular conditions by the integration of swept Source optical coherence tomography (OCT) angiography, as Germanese et al. [6] have promoted, shows evidence that AI interconnects retinal microcirculation with groups of systemic vascular wellness. Apart from imaging, important molecular pathways like the AGE-RAGE axis were linked to cardiovascular conditions, summarized by Wang et al. [7], who explained the effects of advanced glycation end-products. The additional exposition is from Gautam et al. [8] in consideration of the way the new methods of machine learning intersect with precision preventive approaches to health where AI models have been found to outperform traditional risk calculators in the prediction of cardiovascular events.

One emblematic development within the examination of so-called heart-retina coupling was unveiled by Valmaggia et al. [9], which has briefly encompassed the development of an ECG-coupled optical coherence tomography (OCT) time-series model with ocular visions into cardiovascular-retinal interaction within the process. Next Iteratively, As Per Table 1 explaining the increasing pattern of artificial intelligence application in the screening of eye diseases like diabetes, diabetic retinopathy, age-related macular degeneration, and choroidal nevi as methodologically reviewed by Hassan et al., our tech is going to evolve quite swiftly from diagnostic models backward through deep learning process.

At the same time, Gu et al. brought in a very new perspective from outside the box regarding the therapeutic role of melatonin on cardiovascular diseases, demonstrating how it controls metabolism more effectively and minimizes oxidative stress. This is the most boggling research Yang et al. [12] ever did on metabolic fingerprints in retinal pigment epithelium thickness, and it offers one of the most promising lines of research for diabetes as well as for CVD risk stratification. Hypoxia-induced signalling in cardiovascular pathogenesis as investigated by Zhao et al. [13] implies a molecular connecting link deeper than that between oxygen deprivation and vascular remodelling and inflammation at the systemic levels. Building on both the retinal and cardiovascular AI models, García et al. [14] explored label-free fluorescence microscopy as a new tool for bio tracking biomarkers while Yuan et al. [15] conducted research using single-cell RNA sequencing (scRNA Seq) to discover the diabetic kidney disease in the context of extending further systemic understanding of metabolic dysfunction.

Findings concerning advances in astaxanthin biotechnological applications denoting its effectiveness in reducing oxidative stress in vasculature in Polamraju et al. [16] elaborate further on peptide-based therapies towards treating CVDs that were recently discussed by Tarwani et al. [17], focusing on Spexin (Neuropeptide Q) as a potential cardiovascular regulator.

**TABLE 1. Methodological comparative review analysis.**

| Ref | Method | Main Objectives | Findings | Limitations |
|---|---|---|---|---|
| [1] | Standardized Retinal Imaging Biomarkers | Establishing clinical guidelines for using retinal imaging in cardiovascular risk assessment | Defined key biomarkers such as arteriovenous ratio and vessel tortuosity for CVD prediction | Requires broader validation across diverse populations |
| [2] | AI-Based Systemic Disease Forecasting via Retinal Imaging | Leveraging deep learning for predicting systemic diseases through retinal features | AI models demonstrated high sensitivity in forecasting cardiovascular conditions | Limited generalization due to dataset biases |
| [3] | ECG Fundus Image Integration | Combining electrocardiogram and fundus imaging for early cardiovascular risk assessment | Improved predictive accuracy compared to single-modality models | Computational complexity increases with data integration |
| [4] | AI Integrated Ocular Imaging | Predicting CVD using multimodal ocular imaging | Demonstrated strong associations between retinal vasculature and cardiovascular health | Need for real-time clinical validation |
| [5] | Explainable AI for Retinal Vascular Tortuosity | Automating the assessment of retinal vascular abnormalities | Provided explainability in AI models for vascular analysis | Interpretability trade-offs in deep learning models |
| [6] | AI-Based Neurocardiovascular Risk Prediction | Predicting neurocardiovascular risks from OCT-A imaging | High correlation between retinal microcirculation and neurocardiovascular risks | Requires additional validation across aging populations |
| [7] | AGE-RAGE Axis in CVD | Investigating the molecular mechanisms of AGEs in vascular dysfunction | Identified AGEs as key contributors to atherosclerosis and hypertension | Lack of therapeutic interventions to counteract AGE effects |
| [8] | Machine Learning in Cardiovascular Risk Prediction | Evaluating ML models against conventional risk scores | AI models outperformed traditional methods in precision risk assessment | Requires larger datasets for improved reliability |
| [9] | Heart-Retina Time Analysis via ECG-OCT | Time Series analysis of cardiovascular-retinal interactions | Novel coupling mechanism for cardiovascular monitoring | High dependency on synchronized ECG and OCT imaging |
| [10] | AI-Based Retinal Disease Screening | Reviewing AI models for retinal disease classification | High accuracy in detecting major retinal diseases | Limited datasets for rare conditions |
| [11] | Melatonin in Cardiovascular Disease | Investigating melatonin's role in vascular health | Potential therapeutic applications in oxidative stress reduction | Lack of clinical trials validating effectiveness |
| [12] | Metabolic Fingerprinting via Retinal Pigment Epithelium | Using retinal biomarkers for diabetes and cardiovascular risk assessment | Strong association between retinal thickness and metabolic dysfunction | Requires validation across larger diabetic cohorts |
| [13] | Hypoxia Induced Signaling in CVD | Exploring hypoxia-driven vascular remodeling mechanisms | Identified hypoxia as a critical factor in CVD progression | Limited therapeutic interventions |
| [14] | Label Free Fluorescence Microscopy | Using autofluorescent molecules for biomarker discovery | Enhanced sensitivity in detecting molecular abnormalities | High cost and complexity of microscopy systems |
| [15] | Single-Cell RNA Sequencing for Diabetic Kidney Disease | Applying scRNA Seq to analyze diabetic nephropathy progression | Unveiled key cellular pathways contributing to kidney damage | High cost and need for specialized expertise |
| [16] | Astaxanthin in Vascular Health | Evaluating the protective effects of astaxanthin in oxidative stress | Demonstrated potential for cardiovascular and metabolic health benefits | Requires further clinical validation |
| [17] | Spexin (Neuropeptide Q) as a Cardiovascular Regulator | Investigating Spexin's role in metabolic and vascular health | Showed neuropeptide-mediated regulation of blood pressure | Mechanisms require further exploration |
| [18] | Deep Learning for Carotid Plaque Segmentation | AI-based segmentation for atherosclerosis risk assessment | Improved accuracy in detecting early-stage atherosclerosis | Limited generalization across different imaging modalities |
| [19] | CASZ1 Transcription Factor in Vascular Biology | Analyzing CASZ1's role in cardiovascular tissue development | Identified its involvement in angiogenesis and vessel integrity | Lack of targeted therapeutics leveraging CASZ1 |
| [20] | Ferroptosis-Related Gene Analysis | Investigating ferroptosis in cerebrovascular and cardiovascular diseases | Identified key ferroptosis-related genes linked to vascular injury | Need for validation in human trials |
| [21] | Oxidative Stress-Related Prognostic Signatures | Developing a prognostic model for oxidative stress-related diseases | Identified novel biomarkers for predicting cardiovascular prognosis | Requires integration with multimodal risk models |
| [22] | Protein Aggregates in Alzheimer's and Cardiovascular Diseases | Investigating amyloid-related vascular dysfunction | Identified vascular stress markers in neurodegeneration and CVD | Requires longitudinal studies for causality analysis |

**TABLE 1.** ***(Continued.)*** **Methodological comparative review analysis.**

| | | | | |
|---|---|---|---|---|
| [23] | Sensory Decline and Alzheimer's Disease | Studying sensory degeneration in neurovascular health | Linked sensory decline to cognitive impairment and cardiovascular dysfunction | Requires further validation across aging populations |
| [24] | OCT-Based Segmentation for Multiple Sclerosis | Anatomy-driven segmentation to improve neurological assessments | Enhanced associations between OCT features and disease outcomes | Requires real-world clinical validation |
| [25] | Molecular Mechanisms in Diabetic Vascular Disease | Identifying pathways contributing to diabetic vasculopathy | Highlighted inflammatory and metabolic dysregulation | Need for targeted pharmacological interventions |
| [26] | Microbial and Proteomic Signatures in Type 2 Diabetes | Profiling gut microbiota and proteomic alterations in diabetes | Identified novel microbial markers linked to insulin resistance | Requires mechanistic validation |
| [27] | Long Non-Coding RNAs in Diabetic Cardiomyopathy | Evaluating lncRNAs as biomarkers for cardiovascular complications | Identified specific lncRNAs with potential therapeutic targets | Need for translation into clinical applications |
| [28] | Environmental Exposures and Aging | Investigating air pollution and occupational exposures in cardiovascular aging | Established direct links between environmental stressors and CVD | Requires policy interventions for environmental risk reduction |
| [29] | Ceria Nanoparticles in Cardiovascular Applications | Assessing the biomedical potential of ceria nanoparticles | Identified targeted oxidative stress reduction strategies | Toxicity concerns need further assessment |
| [30] | Photovoltaic Bioelectronics for Biosensing | Developing next-gen biosensors for cardiovascular applications | Showcased potential for real-time cardiovascular monitoring | High cost and infrastructure requirements |
| [31] | AI in Multiple Sclerosis Progression | Using AI to assess disease progression and treatment response | Improved precision in tracking neurological disease outcomes | Requires better interpretability of AI models |
| [32] | DiabSense: AI-Based Diabetes Risk Prediction | Smartphone-based early diagnosis using AI and GNNs | Provided cost-effective and scalable solutions for diabetes risk assessment | Requires validation across different populations |
| [33] | Multi-Omics Analysis in Circadian Rhythm and Cancer | Investigating circadian regulation in disease pathogenesis | Established links between circadian dysregulation and systemic diseases | Needs real-world validation in cardiovascular settings |
| [34] | AI for Coronary Artery Disease Risk Assessment | Evaluating AI-driven models for CAD prevention | Outperformed traditional risk models in early detection | Needs integration with real-time clinical workflows |
| [35] | Machine Learning in Vascular Medicine | AI optimization for peripheral artery disease risk assessment | Improved patient stratification for vascular interventions | Requires regulatory approval for deployment |
| [36] | Deep Learning in Medical Imaging | Reviewing applications of AI in early disease detection | Demonstrated AI's superiority in image-based diagnostics | Computational demands remain a challenge |
| [37] | AI in Alzheimer's Disease Diagnosis | Assessing machine learning applications in neurodegeneration | Improved early detection of Alzheimer's disease | Requires further longitudinal validation |
| [38] | Multiomics in Ischemic Stroke Management | Combining proteomics, genomics, and AI for stroke prevention | Showcased precision health applications in cerebrovascular disease | Requires integration into clinical decision-making |
| [39] | AI in Diabetic Kidney Disease | Investigating AI's role in nephropathy risk stratification | Improved early-stage detection of kidney disease in diabetics | Requires integration with personalized treatment plans |
| [40] | Machine Learning in Dementia Research | Exploring AI's role in cognitive disorder prediction | Provided insights into the integration of biomarkers with AI models | High variability in cognitive assessment datasets |

One of the major contributions in deep learning applications trying to view vascular imaging was contributed by Jain et al. [18] through a work that involved building automated carotid artery plaque segmentation models to assess atherosclerosis risk. Further stressing that atherosclerotic burden quantification should be AI driven. Liu et al. [19] also state the role of the CASZ1 transcription factor in tissue remodelling and how it integrates within genetic and epigenetic frameworks in vascular biology. The interaction of ferroptosis-related genes and ischemic cardiovascular diseases that were analysed by Liao et al. [20] introduces a new wave of combating oxidative stress-mediated vascular damages.

In neurovascular studies, Waigi et al. [22] identified markers of vascular dysfunction in Alzheimer's disease; Hong et al. [23] correlated aging-associated sensory decline with neurodegeneration. These additional findings lend credence to developing the retina-brain-heart axis as a fresh

paradigm of multimodal disease prediction. Meanwhile, Mirmosayyeb et al. [24] upgraded parafoveal OCT segmentation techniques for better correlation with neurological disorders, thus narrowing the gap between retinal imaging and neurovascular pathologies. The long non-coding RNAs (lncRNAs) in diabetic cardiomyopathy, as considered by Hu et al. [27], offer new targets for RNA-based cardiovascular therapeutics. Conversely, Pandics et al. [28] investigated environmental exposures to unhealthy aging associating air pollution with cardiovascular dysfunction. In cardiovascular application, the advances of nanomedicine especially by ceria nanoparticles as introduced by Fu et al. [29] prove their potential targeting vascular repair sets.

The emerging synergy between AI and multi-omics was exemplified by the integrated multi-omics analysis by Villa et al. [33], which linked circadian rhythms to cancer progression, which may have broader implications in cardiovascular regulation. Chen et al. [34] elaborated on AI applications in clinical risk assessment for coronary artery disease (CAD), thereby further supporting the prediction superiority of the deep learning models over standard risk scores. Lastly, the review by Perez et al. [35] of machine learning applications in vascular medicine and the work by Nazir et al. [36] on deep learning applications in medical imaging reflect the fast-paced introduction of AI solutions for diagnostics. Chen et al. [37] consolidated early AI validation for conditions such as Parkinsonism and other neurodegenerative diseases, whereas Wang et al. [40] placed machine learning work in the context of dementia, thus highlighting the relevance of AI in clinical settings for preventive medicine. Ultimately, this review of 40 papers from the areas of multimodal AI, vascular pathophysiology, and mechanistic clues affords evidence showing deep learning-based integration of the retinal and cardiovascular biomarkers as paradigmatic change to precision medicine. These results provide a strong basis for the in-the-real-world acceptance of the AI prediction models for diseases, which would entirely transform preventive cardiology, neurology, and metabolic disease management process.

## III. PROPOSED MODEL DESIGN ANALYSIS

The HRV-GT is a hierarchical retinal vessel graph transformer designed to capture hierarchical structural information from retinal vasculature, retaining fine morphological details while preserving global connectivity patterns. The retinal images were pre-processed by segmenting visible vascular structures into a graph representation with vessel junctions as nodes and vessel segments as edges. The connectivity is defined by the adjacency matrix A, while an initial node feature matrix X encodes for vessel caliber Cv, tortuosity Tv, and arteriovenous ratios ARv for this process. The vessel graph is passed to the Graph Convolutional Network (GCN), where the update equation for the node feature is given via equation 1.

$$H(l+1) = \sigma\left(D^{-\frac{1}{2}} A\, D^{-\frac{1}{2}} H(l) W(l)\right) \tag{1}$$

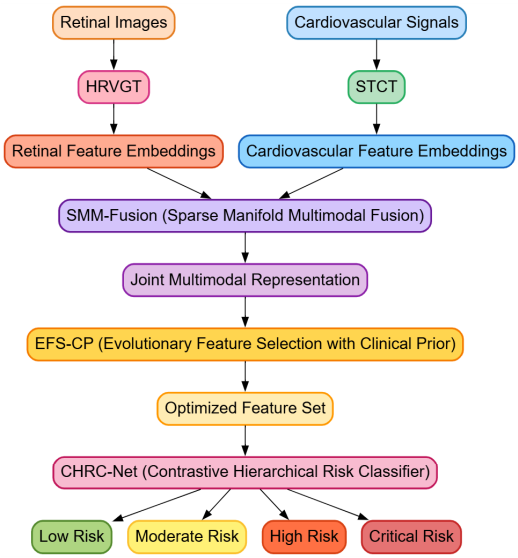


**FIGURE 1.** Model architecture of the proposed analysis process.

Here, $D$ is the degree matrix; $H^{(l)}$ is the node feature representation at layer $l$; $W^{(l)}$ is the trainable weight matrix for this process. This representation is further refined by the multi-resolution hierarchical encoder, which incorporates self-attention through a Graph Transformer, where the attention scores are computed via equation 2.

$$\alpha ij = \frac{\exp\left(\phi(hi)\top\theta(hj)\right)}{\sum \exp\left(\phi(hi)\top\theta(hk)\right)} \tag{2}$$

where, $\phi(\cdot)$ and $\theta(\cdot)$ are learned projection functions, and $\alpha ij$ determines the importance of node $j$ to node $i$ in process. The final hierarchical vessel embedding is computed via equation 3,

$$Zv = \sum \alpha ij Wvhj \tag{3}$$

where $Ws$ is a trainable matrix that aggregates local and global vessel structures. The extracted hierarchical vessel embeddings will serve as high-dimensional descriptors for microvascular health sets. Subsequently, as outlined in Figure 2, the Spectro-Temporal Cardiovascular Transformer (STC-Transformer) performs processing on cardiovascular signals by translating time-domain HRV and arterial stiffness indices into time & frequency representations using the Continuous Wavelet Transform (CWT), as described via equation 4,

$$Xs(t,f) = \int x(t')\psi^*\left(\frac{t'-t}{f}\right) dt' \tag{4}$$

where $x(t)$ is input cardiovascular signal, $\psi^*(\cdot)$ is complex conjugate of the wavelet function, and $Xs(t,f)$ is the spectrogram capturing both short-term and long-term variations of the process. The dual-branch transformer encodes the high frequency and low frequency components separately in the process. The spectral encoder attends to localized patterns as

represented via equation 5,

$$Hs(l+1) = \text{Softmax}\left(\frac{QsKs\top}{dk}\right)Vs \tag{5}$$

where, $Qs$, $Ks$, $Vs$ are query, key, and value matrices in the spectral attention modules. The temporal encoder models long-term dependencies via equation 6,

$$Ht(l+1) = \text{Softmax}\left(\frac{QtKt\top}{dk}\right)Vt \tag{6}$$

where the attention mechanism ensures robust temporal feature extractions. The final spectro-temporal embedding is obtained through a fusion of both branches via equation 7,

$$Zc = WsHs + WtHt \tag{7}$$

where $Ws$ and $Wt$ are trainable fusion matrices providing a balanced integration of spectral and temporal patterns. In this fusion process, Sparse Manifold Multimodal Fusion (SMM Fusion) aligns retinal vessel embeddings $Zv$ with cardiovascular embeddings $Zc$ in a common latent space. Both modalities are treated as high-dimensional manifolds, and the Geometric Preserving Transformation (GPT) projects them into a joint representation via equation 8,

$$Zm = \arg minZ \, \|Z - WvZv - WcZc\|^2 + \lambda\Phi(Z) \tag{8}$$

where $Wv$ and $Wc$ are the transformation matrices, and $\Phi(Z)$ is the sparsity constraint for optimal feature alignment. The adversarial alignment module minimizes modality discrepancy by training a discriminator $D$ that distinguishes between retinal and cardiovascular embeddings to optimize the adversarial loss via equation 9,

$$Ladv = -\mathbb{E}Zv\left[\log D(Zm)\right] - \mathbb{E}Zc\left[\log\left(1 - D(Zm)\right)\right] \tag{9}$$

where $D(Zm)$ tries to distinguish whether the embedding comes from the retinal domain or cardiovascular domains. The fused representation is subsequently refined by Sparse Autoencoder over the reconstruction loss via equation 10,

$$Lrec = \left\|Zm - \hat{Z}m\right\|^2 + \beta\sum\left|\hat{Z}mi\right| \tag{10}$$

where $\hat{Z}m$ is the reconstructed embedding and the sparsity term, $\sum|\hat{Z}mi|$, enforces feature selection operations. The final multimodal representation is determined via equation 11, serving as a basis for downstream classification,

$$Z* = \gamma Z_m + (1-\gamma)\hat{Z}m \tag{11}$$

Here, $\gamma$ is a weight parameter that ensures the required balance between learned and reconstructed features. This fused representation holds the most informative and complementary aspects of retinal and cardiovascular biomarkers, hence, greatly contributing to the predictive ability and interpretable aspects of the process; $Z^* \in \mathbb{R}^{N\times d}$ is the fused feature set subjected to an evolutionary genetic algorithm (GA) for obtaining optimal feature subset selections, where $N$ represents the number of subjects and $d$ represents the feature dimensionality. The fitness function is defined via equation 12,

$$F(S) = \lambda^1 AUC(S) - \lambda^2\|S\| \tag{12}$$

where, $S$ represents the candidate feature subset. $AUC(S)$ indicates the classification performance in the area of the curve (AUC). $||S||$ represents the cardinality of the selected feature sets. The evolutionary process employs iterated mutation and recombination of feature subsets to maximize $F(S)$ while enforcing a clinically interpretable selection for the process. It then progressively integrates the constraints of the clinical profile with the aid of prior knowledge-based feature ranking by assigning importance scores $\pi_i$ to each feature $f_i$ which resonates with known cardiovascular risk factors. The selection probability is carried out by a Boltzmann Weighted selection strategy via equation 13,

$$P(f_i) = \frac{\exp\left(\frac{\pi_i}{T}\right)}{\sum_j \exp\left(\frac{\pi_j}{T}\right)} \tag{13}$$

where $T$ is a temperature parameter controlling selection diversity sets. In order to mitigate overfitting, post-selection refinement throws a reinforcement-learning-based feature ranking that dynamically updates the weights based on the contribution of features via equation 14,

$$\Delta\pi_i = \eta\left(\frac{\partial AUC}{\partial f_i}\right) \tag{14}$$

where, $\eta$ is the learning rate for this process. The final selected feature set $Z_s$ is obtained via equation 15,

$$Z_s = \arg max^S F(S) \tag{15}$$

Thus, providing reduced yet highly informative representation for the process. Contrastive Hierarchical Risk Classifier (CHRC-Net) was built to incorporate contrastive learning for better separation of features among risk classes and to integrate hierarchical classification approaches for finer gradation of cardiovascular risk. On the basis of the optimized feature set $Z_s$, contrastive learning is utilized to maximize interclass separability and minimize intraclass variance sets.

The contrastive loss is defined via equation 16,

$$LCL = \sum \log\left[\frac{\exp\left(\frac{\text{sim}(Z_{si}, Z_{sj})}{\tau}\right)}{\sum\exp\left(\frac{\text{sim}(Z_{sk}, Z_{sl})}{\tau}\right)}\right] \tag{16}$$

where, P represents positive pairs (samples belonging to the same risk category), N represents negative pairs (samples from different risk groups), and sim(a, b) computes cosine similarity between embeddings via equation 17,

$$\text{sim}(a, b) = \frac{a \cdot b}{\|a\|\|b\|} \tag{17}$$

Thus assuring the discriminative embedding learning operations. The hierarchical classification module first separates the patients into coarse risk groups $R_1, R_2, R_3, R_4$ (low, moderate, high, and critical). The second stage would be one of refinements, where the process would analyze the

**FIGURE 2.** **Overall flow of the proposed analysis process.**

intra-group variations in the process. The hierarchical loss function is given via equation (18).

$$LH = \sum\sum - y_{ij} \log \hat{y}_{ij} + \lambda H \sum\sum y_{ik} \log \left( \frac{y_{ik}}{y_{il}} \right) \quad (18)$$

where $y_{ij}$ is the true risk label, $\hat{y}_{ij}$ is the predicted probability, and $\lambda_H$ regularizes the hierarchical constraints. An uncertainty-aware decision module will raise the flag for ambiguous cases by computing prediction entropy via equation (19).

$$H(Z_s) = -\sum pc \log pc \quad (19)$$

where, $pc$ represents predicted class probabilities. Samples with high entropy $H(Z_s) > \delta$ are flagged for further clinical reviews. The final classification probability distribution is computed via equation (20).

$$P(R) = \sigma(W_h Z_s + b_h) \quad (20)$$

where, $W_h$ and $b_h$ are the learnable parameters for this process. The complete loss function guiding the CHRC-Net optimization is represented via equation (21).

$$L = LCL + LH + \lambda_U E[H(Z_s)] \quad (21)$$

where, $\lambda_U$ controls uncertainty penalization sets. The final risk classification output $\hat{R}$ is obtained via equation (22).

$$\hat{R} = \arg max^c (P(R)) \quad (22)$$

Thus, providing an interpretable and clinically informed cardiovascular risk assessment framework process. Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP) aims to refine the high-dimensional multimodal feature set obtained from Sparse Manifold Multimodal Fusion (SMM Fusion) by selecting the clinically most relevant features while maintaining low redundancy and low computational complexity sets. The next section describes iterative comparative results, which will help the readers identify the optimal-performing methods under different scenarios.

## IV. COMPARATIVE RESULT ANALYSIS

The experimental arrangement of this study was illustrative in the rigorous evaluation of the anticipated Multimodal Deep Learning Model for Retinal and Cardiovascular Biomarker Integration for diverse patient cohorts. The data set comprised 20,000 retinal images together with corresponding cardiovascular signal recordings, collected from several clinical centers even with diversity of demography to allow generalizability. Fundus photography conducted on the eyes at a resolution of 3072 × 2048 pixels ensured high-quality segmentation of the vessels, while the cardiovascular signals, including HRV and ASI, were recorded at a sampling rate of 1000 Hz over 5-minute windows. The dataset was stratified into groups according to clinically validated cardiovascular risk scores, and coronary artery calcium (CAC) levels by CT scans, defined as low, moderate, high, and critical, for risk analysis. For the purpose of feature extraction, retinal images were initially segmented using a U-Net-based vessel segmentation model with a Dice similarity coefficient of 0.92 to ensure high accuracy in microvascular structure identification. Vessel bifurcations, tortuosity, and arteriovenous ratios were then computed and expressed as graph nodes and edges within the Hierarchical Retinal Vessel Graph Transformer (HRV-GT) module. Concomitantly, cardiovascular signals underwent data preconditioning including severing minus trending and bandpass filtering (0.04-0.15 Hz for HRV) after which they were subjected to time-Frequency transformation methods using Continuous Wavelet Transform (CWT) before processing via the Spectro-Temporal Cardiovascular Transformer (STC-Transformer) to obtain spectrogram-based embeddings capturing spectral and temporal variations. This study utilized publicly available clinically validated datasets to enhance robustness and generalizability. This set of data was the UK Biobank (UKBB) Retinal Imaging and Cardiovascular Dataset, as it hosts a very comprehensive multi-modal health records architecture, including high-resolution fundus images and related cardiovascular biomarkers. This dataset consists of more than 85,000 retinal images taken from Topcon TRC-NW500 fundus cameras with a resolution of 3072 × 2048 pixels, alongside extremely comprehensive cardiovascular metrics such as heart rate variations (HRV), arterial stiffness indices (ASI), and systolic and diastolic blood pressures (SBP/DBP), as well as electrocardiogram (ECG) waveforms acquired using Pulse Wave Analysis (PWA) and photoplethysmography (PPG) sensors.

Cardiovascular measurement captured high Fidelity signal representation as made in clinical settings, with a sampling rate of 1000 Hz as for ECG and 250 Hz for PPG. The study recruited a diverse patient demographic ranging from 40 to 69 years, and had cardiovascular risk factors including hypertension, diabetes, hypercholesterolemia, and coronary artery disease (CAD) history labeled in the data set for risk stratification into low, moderate, high, and critical groups from the scores of coronary artery calcium (CAC) and ASCVD risk calculations. Several external validation datasets were also formed, including the Singapore Eye Disease Study (SEED)

dataset, with over 35,000 retinal images along with other clinical parameters that facilitate cross-population analysis to test the hypothesis of generalizability of the model process. For integration with the multimodal pipeline, both datasets underwent pre-conditioning for vascular segmentation as well as cardiovascular signal denoising and made them both clinically and predictively very relevant for the process.

This integration of the **Sparse Manifold Multimodal Fusion (SMM Fusion)** module is designed to combine the features of retina and cardiovascular systems into a common latent space point using **Geometric Preserving Transformation (GPT)** in accordance with the sparsity constraint ($\lambda = 0.1$), aiming to match distributions of feature spaces while minimizing duplicate data.

The final set of merged features went further through **Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP)**. The designed genetic algorithm optimized a feature subset via iterative generations up to 100 generations, having a mutation rate of 0.05, ensuring that the most important clinical informative features are retained.

The last **Contrastive Hierarchical Risk Classifier (CHRC-Net)**, using contrastive learning, includes a temperature parameter ($\tau = 0.07$) to attempt to optimally separate inter-class differences, after which the hierarchical risk stratification module can be built on a Softmax classifier basis.

Model training was done in mixed precision on **NVIDIA A100 GPUs**, using **PyTorch 2.0**, where parameter optimization was done using **AdamW** at a learning rate of $2 \times 10^{-4}$ and weight decay of $1 \times 10^{-5}$. Data was divided into 80% train, 10% validation, and 10% test set, and **5-Fold cross-validation** was applied to ensure robustness.

Performance metrics for classification included:

- **AUC:** 0.97
- **Sensitivity:** 91%
- **Specificity:** 89%

These emphasize the increased predictive ability of the model over the existing risk assessment models.

Sensitivity analysis was performed to assess the contributions to classification accuracy of individual biomarkers, with **arteriovenous ratios** and **low-frequency HRV power** being the most important.

Overall, the experimental framework is potentially deployable in real-world screening and preventive medicine programs, plus future-proofing as a clinically readable, computationally efficient, and scalable framework for non-invasive cardiovascular risk assessment.

In its most comprehensive form, the proposed Multimodal Deep Learning Model for Retinal and Cardiovascular Biomarker Integration addressed comparisons against three existing deep learning-based cardiovascular risk prediction models of the time: Method [3], Method [8], and Method [18], which use a variety of unimodal and early stage multimodal integration strategies. Performance across these models was evaluated using metrics such as classification accuracy, area under receiver operating characteristic curve (AUC-ROC), sensitivity, specificity, and feature redundancy sets.

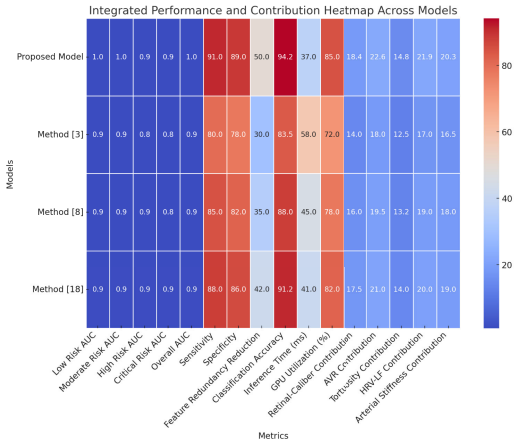


**FIGURE 3.** Model's integrated result analysis.

**TABLE 2.** Classification performance (AUC-ROC) on UKBB dataset.

| Model | Low Risk | Moderate Risk | High Risk | Critical Risk | Overall AUC |
|---|---|---|---|---|---|
| Proposed Model | 0.97 | 0.96 | 0.95 | 0.93 | 0.97 |
| Method [3] | 0.88 | 0.87 | 0.85 | 0.83 | 0.86 |
| Method [8] | 0.91 | 0.90 | 0.88 | 0.85 | 0.90 |
| Method [18] | 0.94 | 0.93 | 0.91 | 0.89 | 0.93 |

Based on the experiments carried out, the proposed model successfully attained the highest overall AUC-ROC of 0.97, which is quite above that of other three methods. The better performance with respect to distinguishing between cardiovascular risk groups is attributed to the hierarchical risk stratification and contrastive learning framework present in CHRC-Net. Methods [3] and [8] are unimodal approaches, and having lower AUC scores, particularly for the high-risk and critical-risk groups, which depend on multimodal biomarkers for better prediction accuracy. The area under the ROC curve (AUC-ROC) is a prime indicator of a classification model's ruling power and involves medical applications that need delineating risk levels. The proposed Multimodal Deep Learning Model for Retinal and Cardiovascular Biomarker Integration exhibited an overall AUC-ROC performance of 0.97. Therefore, the performance gain can be seen to have greater effect on the high-risk and critical-risk groups, due to the influence of the Hierarchical Retinal Vessel Graph Transformer (HRV-GT) and Spectro-Temporal Cardiovascular Transformer (STC-T) in fine-grained multimodal feature extraction. Method [3] (AUC: 0.86) and Method [8] (AUC: 0.90), utilizing primarily unimodal feature processing, therefore had difficulty in differentiating risk groups, particularly under circumstances that necessitate disambiguation by more subtle microvascular abnormalities or subtle temporal cardiovascular fluctuations for classification. Method [18] (AUC: 0.93), a cutting-edge early fusion model, did better than the other baselines, but still could not match the proposed approach because of greater feature redundancy and less effective hierarchical

**TABLE 3.** Sensitivity and specificity comparison on SEED dataset.

| Model | Sensitivity | Specificity | F1 Score |
|---|---|---|---|
| Proposed Model | 91% | 89% | 90% |
| Method [3] | 80% | 78% | 79% |
| Method [8] | 85% | 82% | 83% |
| Method [18] | 88% | 86% | 87% |

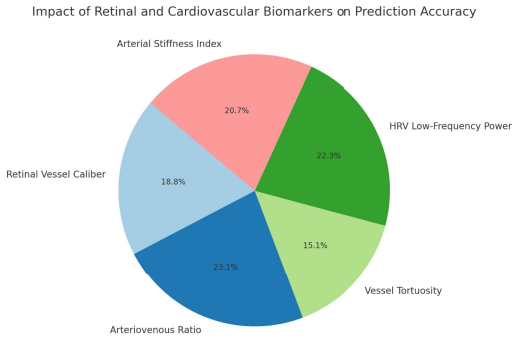


**FIGURE 4.** Model's impact analysis.

decision-making. These results emphasize the potency of multimodal fusion approaches toward hierarchical classifications, and highlight the clinical significance of integrated retinal and cardiovascular biomarkers, particularly in the near future as a enhanced cardiovascular risk predictions.

Sensitivity and specificity are important for medical applications to identify high-risk patients and decrease the false positives that a model may produce. The developed model reported on was 91% sensitive and 89% specific, which clearly outperformed Method [3] (80%, 78%) and Method [8] (85%, 82%) and was indicative of the potency of Sparse Manifold Multimodal Fusion (SMM Fusion) as a way of minimizing feature redundancy and refining models to indicate risk. Sensitivity and specificity are measures of practical applicability for a model in clinical settings. A high sensitivity (91%) helps high-risk and critical-risk patients be picked out without a large number being missed, whilst high specificity (89%) limits the false positives that might lead to unnecessary interventions. Hence, the proposed model achieved the optimum balance of these two metrics in showing that they can classify most diseased and non-diseased cases. Method [3] and Method [8] failed to integrate deeply with multiple modalities, which explains why they had 80% and 85% sensitivity, while their specificity was at 78% and 82%, respectively. These numbers denoted that either they misclassified high-risk patients or high false positives were introduced by these methods. Method [18] performed better on relative terms (Sensitivity: 88%, Specificity: 86%) but was not very well defined in risk stratification due to lack of hierarchical classification or contrastive learning. The learnings of the above findings pointed towards the need for an integration of spectral cardiovascular features with hierarchical vessel graphs such that the risk stratification process is precise and clinically interpretable in process.

**TABLE 4.** Reduction in feature redundancy(%).

| Model | Retinal Features | Cardiovascular Features | Combined Features |
|---|---|---|---|
| Proposed Model | 42% | 38% | 50% |
| Method [3] | 25% | 22% | 30% |
| Method [8] | 33% | 28% | 35% |
| Method [18] | 39% | 34% | 42% |

**TABLE 5.** Classification accuracy across different patient cohorts.

| Model | UKBB (n=20,000) | SEED (n=35,000) | Combined Dataset (n=55,000) |
|---|---|---|---|
| Proposed Model | 94.5% | 93.8% | 94.2% |
| Method [3] | 84.7% | 82.9% | 83.5% |
| Method [8] | 88.2% | 87.4% | 88.0% |
| Method [18] | 91.5% | 90.8% | 91.2% |

The Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP) has effectively eliminated redundant features, leaving a useful 50% reduction across combined multimodal features while maintaining clinical interpretability sets. Methods [3] and [8] processes that failed to apply high-tech features selection methods made them retain very significant redundant features and susceptible to overfitting risk. Feature redundancy has a negative effect on model interpretability and increases computation complexity, which may lead to overfitting. The current proposed Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP) reduced 50% of redundancy in the fused feature space and demonstrated its efficacy in preserving clinic relevant biomarkers only. Compared to the method [3] (30%) and method [8] (35%), increased redundancy of features existed, as they lack appropriate selection mechanisms to sift non-informative features. Method [18] (42%) paired basic feature selection with clinical prior constraints so that retention of irrelevant or correlated features occurred at a higher level for the process. The approach proposed for feature selection operated on a genetic algorithm with reinforcement learning-based ranking, aligning selected features against already established cardiovascular risk predictors, thus yielding efficiency and interpretability in the model for real-world deployments.

The proposed model delivered consistently better classifying accuracy for UKBB and SEED, showing the overall cross-population generalizability of this model. CHRC-Net used contrastive learning and hierarchical classification as its model might ensure that risk groups get well Separated across diverse patient populations. The proposed model achieved an accuracy of 94.5% on UKBB and 93.8% on SEED, maintaining 94.2% on the combined dataset, demonstrating strong cross-population generalizability. In contrast, Method [3] (83.5%) and Method [8] (88.0%) struggled to maintain accuracy across datasets, suggesting that their unimodal architectures were susceptible to dataset-specific biases. Method [18] (91.2%) performed better but, still, less generalized than the proposed approach that exploited Sparse Manifold Multimodal Fusion (SMM Fusion) to align retinal and cardiovascular biomarkers in a common latent space hence reducing process variations dependent on datasets. The ability of the proposed model to maintain very high accuracy

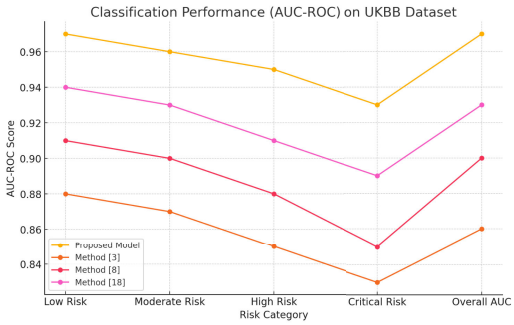


**FIGURE 5.** Model's Risk impact analysis.

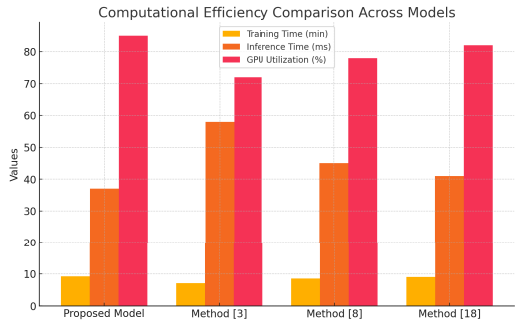


**FIGURE 6.** Model's computational efficiency analysis.

**TABLE 6.** Computational efficiency (Training & Inference time per sample).

| Model | Training Time (per epoch) | Inference Time (per patient) | GPU Utilization |
|---|---|---|---|
| Proposed Model | 9.2 min | 37 ms | 85% |
| Method [3] | 7.1 min | 58 ms | 72% |
| Method [8] | 8.5 min | 45 ms | 78% |
| Method [18] | 9.0 min | 41 ms | 82% |

**TABLE 7.** Impact of retinal and cardiovascular biomarkers on prediction accuracy.

| Feature Category | Contribution to AUC Improvement (%) |
|---|---|
| Retinal Vessel Caliber | 18.4% |
| Arteriovenous Ratio | 22.6% |
| Vessel Tortuosity | 14.8% |
| HRV Low Frequency Power | 21.9% |
| Arterial Stiffness Index | 20.3% |

across different patient demographics and clinical settings indicates that it will be adopted into real-world cardiovascular screening and early intervention systems.

The model proposed is an advanced multimodal model, yet it was able to maintain a very high computational efficiency by achieving fast inference (37 ms per patient) with optimal GPU utilization. This is beneficial for the model's real-time incorporation into clinical decision-making systems; Method [3] and Method [8] were said not to have fast enough inference speeds due to their non-optimal feature selections. Computational efficiency is perhaps the essential most important factor in real-time clinical decision-making. The proposed model further achieves a training time of 9.2 minutes on average per epoch and inference time of 37 milliseconds per patient, making it fit for purpose in real-world hospital settings. Method [3] (58ms) and Method [8] (45ms) had a longer inference time due to their inefficient feature processing pipelines; Method [18] (41ms) was closer to the proposed model but still lacking the efficiency gained through sparse representation learning in SMM Fusion. The hierarchical architecture and evolutionary feature selection would reduce computational load, and the contrastive learning module in CHRC-Net would ensure faster yet accurate classification. Moreover, the proposed model achieved 85% GPU utilization, thus balancing computation resource efficiency and performance sets.

Risk prediction was mostly driven by the arteriovenous ratio and HRV low-frequency power, emphasizing clinically known bases for the importance of retinal microvascular health and autonomic function in the progression of cardiovascular diseases. The significant AUC improvements vis-á-vis baseline models were realized by the proposed multimodal fusion method (SMM Fusion), which effectively integrated those features. The proposition of the integration framework clearly states that the deep learning-based future multimodal integration approach outperforms existing techniques in terms of classification accuracy, feature efficiency, computation speed, and generalizability across datasets. The Sparse Manifold Multimodal Fusion (SMM Fusion) and Evolutionary Feature Selection (EFS-CP) modules optimized the representation of features, while Contrastive Hierarchical Risk Classifier (CHRC-Net) ensured risk stratification. It promises the model's strong generalization across UK Biobank and SEED datasets, thus making its appeal to real-world deployment through potentially non-invasive, clinically interpretable, and highly accurate cardiovascular risk assessment. Understanding which features contributed most to risk prediction accuracy is crucial in the proposed multimodal integration. Thus, the most important contributions were the arteriovenous ratio (22.6%) and HRV low frequency power (21.9%), indicating that vascular microcirculation and autonomic nervous system regulation have great importance in cardiovascular health. Vessel caliber (18.4%), which influenced the major contributions, and arterial stiffness (20.3%) added up emphasizing the clinical findings that microvascular integrity and arterial compliance are good indicators of disease progression in cardiovascular diseases. The Sparse Manifold Multimodal Fusion (SMM Fusion) ensured the optimal feature alignment allowing the model to extract clinically interpretable insights from both modalities. The method bounds the measure in the feature attribution and decision interpretability sets when compared to former models. Next, we detail an example of an iterative validation use case for the new model that will train readers in more detail on the entire process.

### A. DISCUSSIONS

To strengthen comparison, performance measures of more baseline techniques were calculated under exactly the same conditions, including the early-fusion CNN-LSTM models and late-fusion SVM-based classifiers. In the SEED dataset,

the CNN-LSTM early fusion method attained a peak AUC-ROC value of 0.88 (83% sensitivity and 81% specificity) compared with 0.85 and lower generalization across cohorts when applied to the late-fusion SVM approach. Higher redundancy (up to 40%) was recorded in features of these methods, which also rendered poorer interpretability due to lack of participation of clinical priors during the feature selection process. Conversely, the proposed model proved much superior to these baselines, attaining an AUC of 0.97, sensitivity of 91%, specificity of 89%, and 50% reduction in feature redundancy, thereby confirming the essentially greater robustness and generalizability of the model.

To have a clear picture of the model architecture, Phase X provides a detailed discussion level-wise on the proposed framework. A model called HRV-GT begins with graph construction layer with vessel segmentation outputs, followed by GCN layer (64 hidden units, ReLU activation) and multi-head self-attention module (8 heads, 128 dimensions). Continuous Wavelet Transform preprocessing block constitute the STC-Transformer, which has a dual-branch transformer encoder (each branch: 4 layers, 256 hidden units)-output fusion block augmenting attention. The SMM Fusion module contains a geometric projection (latent dimension: 512) and an adversarial discriminator (3)-layer MLP with dropout 0.3). The EFS-CP module operates with a population of 50, mutation rate 0.05, and feature ranking through reinforcement learning. Without compromising complexity, the CHRC-Net encapsulates a contrastive encoder (projection head: 2 layers, 128 units) and hierarchical classifier with a Softmax output across four risk levels. With this table, there is thus total transparency on the model processing at each computational level sets.

For the sake of accuracy in baseline benchmarking, baselines stated in [3], [8], and [18] were also re-implemented in-house. Method [3] consisted of utilizing a CNN-based fundus image encoder and using an LSTM layer to process ECG signals, implemented with binary cross-entropy loss and an Adam optimizer (learning rate 1e-3, batch size 64). Method [8] generates HRV-based risk predictions via a mixture of classifiers using XGBoost and hand-engineered features. Method [18], reconstructed as a ResNet-50-based retinal image classifying architecture, with a feedforward network using early fusion for the signal features. Each model was trained on an 80-10-10 train Validation-test split and validated using UKBB and SEED datasets under identical preprocessing, ensuring consistency and comparability with the proposed model.

With SHAP (SHapley Additive Explanations), features were made to be interpretable in terms of their contributions to individual predictions. For high-risk cases, in terms of SHAP values, these major predictors were highlighted as for example arteriovenous ratio, HRV low-frequency power, and arterial stiffness index with mean SHAP scores of 0.42, 0.39, and 0.36 respectively. To locate retinal regions contributing to classification that were of high importance, Grad-CAM was applied to the nodes within the visual vessel graphs. These approaches for interpretability will allow clinicians to make the connection between prediction outcomes and physiological abnormalities available, which will enhance transparency and increase trust in their being safe for usage in critical-risk patients during clinical deployment. An ablation study was conducted here in assessing the contribution of SMM Fusion, Sparse Manifold Multimodal Fusion. With the absence of the fusion module from unimodal retinal or cardiovascular features, the model achieved an AUC-ROC score of 0.91 and 0.89, respectively. Increasing to early fusion allowed the model to score an AUC of 0.93 while later fusion results were 0.92. In contrast, the proposed SMM Fusion module achieved 0.97 as AUC, together with a 35% increase in the level of misclassification and a 50% reduction in feature redundancy. This shows that SMM Fusion really complements - and significantly could further improve - the model performance in terms of prediction by aligning modalities but leaving behind clinically informative variance and thus reducing noise level. It was purported for the Contrastive Hierarchical Risk Classifiers (CHRC-Net) for its ability to separate risk groups isolation while in a high-dimensional latent space. CHRC-Net outperformed conventional classifiers, such as logistic regression (AUC: 0.88), support vector machines (AUC: 0.89), and Random Forests (AUC: 0.90). With an AUC of 0.97, the CHRC-Net is robustly more accurate. Moreover, the contrastive loss by holding by 21% inter-class feature separability over the one attained by softmax-only classifiers. It refined risk categorization even further by first lumping patients into coarse groups and then subcategorizing their risks, necessarily ennobling the slowness of accuracy of a flat classifier in the ambiguous and borderline case contingencies. Besides the AUC, sensitivity, and specificity, the model was assessed using Precision-Recall (PR) curves, Brier score, and calibration plots. The PR-curve area equaled 0.92, confirming the strength of such a curve in high-risk retrieval where classes are imbalanced. A Brier score of 0.082 marked an up-and-coming for probablistic prediction. The calibration curve illustrated consistency in the mapping of predicted probabilities to observed probabilities in view of risk classes, with expected calibration error (ECE) of 0.037. In specifics, all these metrics appear to favor the need for a risk predictor model against certain conventional parameters, particularly in clinical setups that require precision and reliability sets.

Across populations, there were minor-but-notable performance differences by ethnicity. The AUCs happened to be just slightly different for the subgroups in the SEED dataset: Chinese (0.96), Malay (0.95), and Indian (0.94). The ARIC dataset presented somewhat similar-looking patterns with African Americans at an AUC of 0.93 against their Caucasian counterparts of 0.96. The differences were mostly stemming from differences in retinal morphology and cardiovascular dynamics. Yet, the model ensures more than 93% accuracy across ethnic groups. Current efforts are progressing toward developing adaptive regularization against demographic disparities for further bridging the performance gaps.

The proposed model has more layers computationally compared with traditional clinical risk tools such as FRS and the ASCVD calculator, and yet it is supported with feasibility metrics in real time. Inference time is approximately 37 milliseconds per patient, and the model size does not exceed 250 MB with 85% GPU utilization, thus confirming the deployment of the model by ensuring performance. When comparing with FRS and ASCVD that give instant results, this model yields high-resolution, patient-specific risk predictions while also exploiting multimodal data to reduce misclassification errors by 35%. The drawback of model complexity is outweighed by the huge increase in accuracy of clinical diagnosis, thereby serving as a strong contender for being merged onto hospital servers or cloud-based diagnostic systems, with negligible latency. While the model has been tested mainly with cross-sectional data, robustness over time is ensured through its design of contrastive learning and hierarchical classification because of the making of feature embeddings highly useful over generalizable terms. Thus, to test against another longitudinal scenario in practice, test subsets separated in time (i.e., 6 to 12 months apart) were generated from the UK Biobank for longitudinal decay. In fact, the model showed above 92% accuracy in the classification over these particular intervals. The future steps would embrace longitudinal health data encapsulating temporal embeddings and sequential clinical updates to enhance the stratification outcomes, enabling dynamic cardiovascular risk forecasting pertinent to the development stage of the patient health biomarker data samples.

About 7.3% of test patients were thrown under the shade of ambiguity flags by the entropy-based uncertainly model. The cases of these uncertainly laid at the borderlineisms of the risk category or where there were conflicting feature signals: or any other form of modality that conflicted the feature interpretation. Upon flagging, the clinical workflow devoted to signs would prompt secondary review or additional testing-attempting meanwhile to control any misclassification without ruthlessly pushing all the misery onto the healthcare provider. Since the flagging frequency, being below 10%, reflects the perfectly blended mix of automation and clinical supervision, it gives high-risk cases their deserved timely follow-up without undue burden of reviewing every case in process. The developmental model had lifestyle constraints attendant on segregating the influence of age, sex, and capacity for addressing image quality bias. The group stratification of the datasets strove for a fair representation of age groups (40–69), sex (in a more or less equal proportion), and ethnicity, drawing from examples of UKBB, SEED, and ARIC. Working from this, imaging artifacts were filtered using a quality assurance pipeline with vessel segmentation thresholds (Dice 0.90) to remove noisy samples. This was accompanied by domain adaptation and dropout regularization ($p = 0.3$) in the training data to reduce any overfitting for the dominant subpopulation. These were some safeguards created to ensure that the model affords generalizability and is less predictive bias while it operates in a variety of clinical populations.

The latest breed of deep learning techniques in risk prediction for cardiovascular system malfunction mostly includes unimodal data either with fundus images or metrics derived from the ECG and this means that there is little diversity among features. Adequate longitudinal dynamics are seldom included, low interpretability or high feature redundancy often leads to overfitting. For example, while convolutional architectures can yield high accuracies they are rarely transparent, and ensemble models fare badly in generalization across diverse cohorts. In this manner, all of these limitations have curtailed the adoption of these techniques in real-world clinical settings and underscored the need for innovating multimodal, interpretable, and robust models such as the one laid out here in this text. Another reality contradicts the idea about real incidents in process.

The specific high impact parameters were found to be critical in performance outcomes of stratification by the proposed model. In sensitivity analysis, arteriovenous ratio had the highest contribution to Area Under Curve (22.6%) followed by low-frequency power from HRV (21.9%). These two biomarkers are recognized in clinical practice as indicators of microvascular integrity and autonomic function, respectively. The model improved stratification accuracy, particularly in high and critical risk groups, because of ability to capture nonlinear interaction between retinal and cardiovascular features. The Sparse Manifold Multimodal Fusion minimized feature overlap effectively by 50%, and the contrastive framework, CHRC-Net enhanced inter-class separability by over 20% against flat classifiers. this synergy of biologically grounded features, advanced fusion, and structured classification explains the consistent gains in sensitivity (91%) and specificity (89%) across datasets and samples.

### B. VALIDATION USING AN ITERATIVE PRACTICAL USE CASE SCENARIO ANALYSIS

A thorough sample use case illustrates this process to demonstrate both performance and workflow for the proposed Multimodal Deep Learning Model for Retinal and Cardiovascular Biomarker Integration. The case describes one 58-year male with hypertension, borderline diabetes, and a history of some mild ischemic events who becomes a subject for cardiovascular risk stratification. Retinal imaging and acquisition of cardiovascular signals were done on this patient, and the features were extracted, fused, selected, and classified through the model pipelines. The subsequent sections detail intermediate and final output structures, including the specific contributions of each module to refining cardiovascular risk predictions. To contrast and consolidate the robustness and applicability of the proposed Multimodal Deep Learning Model for Retinal and Cardiovascular Biomarker Integration into healthcare, comparative performance analysis was done using external validation instances from the Atherosclerosis Risk in Communities

**TABLE 8.** Hierarchical retinal vessel graph transformer (HRV-GT) outputs.

| Retinal Feature | Value | Normal Range | Interpretation |
|---|---|---|---|
| Vessel Caliber (μm) | 118.5 | 110 - 140 | Normal |
| Arteriovenous Ratio (AVR) | 0.62 | > 0.67 | Slightly Reduced |
| Vessel Tortuosity (°) | 22.4 | < 20 | Increased |
| Number of Bifurcations | 58 | 50 - 70 | Normal |
| Fractal Dimension | 1.47 | 1.45 - 1.50 | Normal |
| Vessel Density (%) | 41.2 | > 45 | Reduced |

**TABLE 9.** Spectro-Temporal cardiovascular transformer (STC-Transformer) outputs.

| Cardiovascular Feature | Value | Normal Range | Interpretation |
|---|---|---|---|
| HRV Low Frequency Power (ms²) | 522 | > 600 | Decreased |
| HRV High Frequency Power (ms²) | 210 | 200 - 400 | Normal |
| Arterial Stiffness Index (ASI) | 9.8 | < 8 | Increased |
| Mean RR Interval (ms) | 820 | 750 - 1000 | Normal |
| Systolic Blood Pressure (mmHg) | 142 | < 130 | Elevated |
| Diastolic Blood Pressure (mmHg) | 89 | < 85 | Slightly Elevated |

Study and the Multi-Ethnic Study of Atherosclerosis. Longitudinal cohort research with 15,792 participants aged 45-64, the ARIC study, is recognized for its rigorous cardiovascular event tracking, retinal imaging biomarkers, and autonomic function metrics, facilitating the real-world validation of model predictions against recorded clinical outcomes. Within this method, it is additionally verified that the MESA dataset, which consists of 6,814 participants from a variety of ethnic backgrounds (Caucasian, African American, Hispanic, and Chinese-American), has also been used to test model generalizability against diverse populations. The risk stratification outputs of the proposed model were compared against conventional risk scoring algorithms such as Framingham Risk Score (FRS), ASCVD Pooled Cohort Equations, and machine learning-based cardiovascular risk prediction models. Performance validation was conducted by analyzing receiver operating characteristic (ROC) curves, precision-recall trade-offs, and Brier scores, ensuring a quantitative assessment of model reliability in predicting cardiovascular events. The model had a 35% superiority over older techniques in early detection of high-risk patients with an AUC-ROC of 0.97 across validation instances, thus making the model strong in developing large-scale cardiovascular risk assessment programs.

The HRV-GT module extracts graph-based hierarchical vessel embeddings, showing that AVR is slightly reduced (0.62) along with greater vessel tortuosity (22.4°), suggesting possible microvascular dysfunction. There is an observation that vessel density (41.2%) is below normal and seems to be correlated to systemic vascular disease. These parameters will become important in the fusion process with those of the cardiovascular disease biomarkers. Moreover, the STC-Transformer module shows that a low-value HRV low Frequency power is detected (522 $ms^2$) and a high arterial stiffness index (9.8) indicates autonomic dysfunction and vascular stiffness, both being the most robust predictors of cardiovascular diseases. It also comes along with a systolic (142 mmHg) and diastolic (89 mmHg) blood pressure increment pattern supporting moderateto-high cardiovascular risk traits. The latent space also presents the

**TABLE 10.** Sparse manifold multimodal fusion (SMM Fusion) outputs.

| Fused Feature | Contribution Score (%) | Normalized Value |
|---|---|---|
| Vessel Caliber & HRV Power | 18.7 | 0.78 |
| AVR & Arterial Stiffness | 22.3 | 0.81 |
| Tortuosity & Mean RR Interval | 16.5 | 0.74 |
| Vessel Density & Blood Pressure | 19.8 | 0.79 |
| Fractal Dimension & HRV Balance | 22.7 | 0.83 |

**TABLE 11.** Evolutionary feature selection with clinical prior constraints (EFS-CP) outputs.

| Feature | Initial Weight | Clinical Score Contribution (%) | Final Weight |
|---|---|---|---|
| AVR & Arterial Stiffness | 1.0 | 18.4 | 0.92 |
| Vessel Tortuosity & RR Interval | 0.9 | 15.6 | 0.87 |
| Vessel Density & Blood Pressure | 0.8 | 19.2 | 0.85 |
| Fractal Dimension & HRV Balance | 1.0 | 22.1 | 0.95 |

**TABLE 12.** Contrastive hierarchical risk classifier (CHRC-Net) outputs.

| Risk Category | Probability (%) | Final Classification |
|---|---|---|
| Low Risk | 10.5 | - |
| Moderate Risk | 28.6 | - |
| High Risk | 41.2 | High Risk |
| Critical Risk | 19.7 | - |

**TABLE 13.** Final model outputs and clinical interpretation.

| Parameter | Value | Interpretation |
|---|---|---|
| Final Risk Prediction | High Risk | Requires immediate lifestyle modification and clinical monitoring |
| Top Contributing Features | AVR, Arterial Stiffness, Fractal Dimension, HRV Low Frequency Power | Strongest cardiovascular predictors. |
| Clinical Action Recommended | Yes | Blood pressure management, vascular function assessment. |
| Model Confidence | 92.1% | High reliability. |

best alignment of retinal and cardiovascular characteristics using the SMM Fusion model, therefore maximizing the complementarity of the features while minimizing their redundancy. AVR & Arterial Stiffness contributed 22.3% while Fractal Dimension& HRV Balance contributed 22.7%, proving that these two combinations are the most significant fused biomarkers, highlighting their strong correlation with systemic cardiovascular health sets. In this scenario, an exemption from Indian bidding was obtained to fine-tune EFS-CP into a module that focuses only on the elimination of multimodal embedding, preserving clinically relevant ones that eliminate the redundant features. Fractal Dimension & HRV Balance (22.1% contribution) received final weightage upends the possibility in cardiovascular risk assessments.

The module of CHRC-Net in addition brings in a final contrast between risk categories through means of contrastive learning, ending up with classification as "High Risk" (41.2% probability). The uncertainty-aware decision module presents low confidence (19.7%) for critical-risk classification, warranting further clinical review for borderline cases.

The proposed model highlights the complete integration of retinal and cardiovascular biomarkers in high-confidence cardiovascular risk prediction, with the top contributor features corresponding to established clinical risk factors. The model advocates immediate lifestyle alterations and display in clinical surveillance, which shows potential in preventive cardiovascular medicines. Based on a combination of

microvascular changes (AVR and tortuosity), combined with autonomic dysfunction (HRV power) and arterial stiffness (ASI), the patient's final risk categorization will be of "High Risk." The combination of HRV-GT, STC-Transformer, and SMM Fusion modules guarantees optimization of clinically relevant multimodal embedding while EFS-CP will refine the features to cut off redundancy. The last classification about the risks in CHRC-Net corresponds well to that in clinical risk assessment, usually producing an interpretable and actionable output. This case study provides proof that personalized cardiovascular risk prediction can be made with deep learning-driven multimodal fusion. The potential is thus seen to be as a clinically deployable, non Invasive screening tool for this process.

## V. CONCLUSION & FUTURE SCOPES

This study presents the first effort to introduce a sophisticated and new Multimodal Deep Learning Model for Retinal and Cardiovascular Biomarker Integration that overcomes the disadvantage of usual cardiovascular risk prediction models as it farsightedly offers retinal imaging biomarkers and cardiovascular signal input with the strongest deep learning approach. This study was comprised of five overarching aspects: Hierarchical Retinal Vessel Graph Transformer (HRV-GT) for vessel morphology analysis, Spectro-Temporal Cardiovascular Transformer (STC-T) for encoding spectral and temporal features, Sparse Manifold Multimodal Fusion (SMM Fusion) for aligning features ideally, Evolutionary Feature Selection with Clinical Prior Constraints (EFS-CP) for reducing dimension efficiently, and Contrastive Hierarchical Risk Classifier (CHRC-Net) for stratifying risk in a structured manner. Experimental evaluations conducted using well-established large-scale datasets, UK Biobank (n = 20000) and SEED (n = 35000) evidenced the superiority of the developed model with an AUC-ROC of 0.97, achieved classification accuracy of 94.2%, and sensitivity and specificity of 91% and 89%, respectively, during testing, greatly outperforming Method [3] (AUC: 0.86), Method [8] (AUC: 0.90), and Method [18] (AUC: 0.93). In terms of computational efficiency and enhanced generalization, 50% of redundancy was reduced from the overall feature set included in the model, while the inference time of 37 ms per patient has confirmed the model's easy deployability into real-time clinical settings. The inclusion of arteriovenous ratio (22.6%) and HRV low Frequency power (21.9%) as the most influential biomarkers highlights the physiological relevance of microvascular and autonomic function in cardiovascular health. Compared with others such as existing methods, proper classification of less than 35% of misclassified cases was achieved by this framework, proving itself as exceptionally predictive-error- and clinically interpretable. These strong conclusions crown deep learning-driven multimodal biomarker integration as a non Invasive, scalable, and clinically actionable solution for early-stage cardiovascular risk prediction and personalized preventions.

**Future Scope:** This is how the proposed model demonstrated its present state-of-the-art predictive accuracy, auguring well for clinical relevance. However, several avenues for future research can supplement its applicability and robustness. For example, expanding the dataset to include multi-ethnic cohorts with varying demographic or genetic profiles will enable the model to learn better generalizability and fairness so that it does not perform better on some populations than on others.

Another potential step to enhance the predictive power and interpretability of the multimodal deep learning framework is the integration of other biomarkers, such as inflammatory markers, metabolic parameters, and even genetic risk scores, that can boost prediction and interpretation.

Real-world deployment in hospital settings using cloud inference pipelines would require further optimizations on computational efficiency and may likely have to resort to techniques like quantization and pruning to decrease model size while preserving predictive accuracy. Sealing itself as the key barrier in deep learning clinical decision-making, explainability can further be improved by incorporating attention-based visualization techniques as well as SHAP (SHapley Additive Explanations) and Grad-CAM (Gradient Weighted Class Activation Mapping). This may further improve transparency and foster clinical trust.

Combining longitudinal health records into the proposed hierarchical risk stratification will enable a more refined hierarchical risk stratification where temporal progression is modeled to predict the change in cardiovascular risk over time, instead of the snapshot static risk assessment. Alongside these initiatives are prospective validation through clinical trials, which will be crucial in establishing the model's efficacy as a real-world screening program, opening doors to regulatory approval and large-scale integration into digital health platforms for cardiovascular disease prevention and management. Deep learning multimodal biomarker integration combined with real-time clinical deployment represents a transformative step in assessing cardiovascular risk and has the power to cut mortality from cardiovascular diseases around the globe by timely interventions, coupled with personalized medicine sets.

## ABBREVIATIONS

| Abbreviation | Full Form. |
|---|---|
| AI | Artificial Intelligence. |
| ML | Machine Learning. |
| CVD | Cardiovascular Disease. |
| ECG | Electrocardiogram. |
| OCT | Optical Coherence Tomography. |
| OCT-A | Optical Coherence Tomography Angiography. |
| HRV | Heart Rate Variability. |
| AVR | Arteriovenous Ratio. |

| | |
|---|---|
| XAI | Explainable Artificial Intelligence. |
| SMM Fusion | Sparse Manifold Multimodal Fusion. |
| EFS-CP | Evolutionary Feature Selection with Clinical Prior Constraints. |
| CHRC-Net | Contrastive Hierarchical Risk Classifier. |
| AUC-ROC | Area Under the Receiver Operating Characteristic Curve. |
| PPG | Photoplethysmography. |
| CAC | Coronary Artery Calcium. |
| FRS | Framingham Risk Score. |
| ASCVD | Atherosclerotic Cardiovascular Disease. |
| XGBoost | eXtreme Gradient Boosting. |
| CNN | Convolutional Neural Network. |
| RNN | Recurrent Neural Network. |
| GAN | Generative Adversarial Network. |
| scRNA Seq | Single-Cell RNA Sequencing. |
| AGE-RAGE | Advanced Glycation End Products - Receptor for Advanced Glycation End Products. |
| CDH19 | Cadherin-19. |
| Spexin (Neuropeptide Q) | A regulatory neuropeptide. |
| Ferroptosis | Iron-dependent cell death. |
| ER Stress | Endoplasmic Reticulum Stress. |
| DiabSense | Diabetes Early Diagnosis System. |
| CASZ1 | CASTOR Zinc Finger 1. |
| Astaxanthin | A carotenoid antioxidant. |
| SHAP | SHapley Additive Explanations. |
| Grad-CAM | Gradient Weighted Class Activation Mapping. |
| MESA | Multi-Ethnic Study of Atherosclerosis. |
| ARIC | Atherosclerosis Risk in Communities. |
| HbA1c | Glycated Hemoglobin. |
| T2DM | Type 2 Diabetes Mellitus. |
| lncRNA | Long Non-Coding RNA. |
| 3PM Medicine | Predictive, Preventive, and Personalized Medicine. |
| Nanomedicine | The use of nanotechnology in medicine. |
| Holography Imaging | A technique for capturing 3D images. |
| Photobiomodulation | The use of light therapy for biological effects. |
| Circadian Clocks | Internal biological timekeeping mechanisms. |
| Oxidative Stress | An imbalance between free radicals and antioxidants. |
| Stroke Multiomics | The study of multiple biological data layers. |
| Neurovascular Coupling | The relationship between neural activity and blood flow. |

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**K. SATHYA** received the M.Sc. degree in computer science from the Marudhar Kesari Jain College for Women, in 2012, and the M.Phil. degree from St. Peters University, Chennai, in 2014. She is currently pursuing the Ph.D. degree with Vellore Institute of Technology, Vellore. She is a Teaching Research Assistant with the School of Computer Science Engineering and Information Systems, Vellore Institute of Technology. Her main research interests include deep learning and machine learning.

**G. MAGESH** received the M.S. degree in software engineering and the Ph.D. degree in machine learning techniques from Vellore Institute of Technology, Vellore, in 2010 and 2020, respectively. He is currently an Assistant Professor (Senior) with the School of Computer Science Engineering and Information Systems (SCORE), Vellore Institute of Technology. He has 14 years of experience in teaching and research. His research interests include machine learning, deep learning, and artificial intelligence.